FIELD GUIDE

10 Data & Ops Workflows I've Automated

…and what each one actually saved.

> *A field guide for EdTech and SMB founders who are tired of their team burning hours on spreadsheets.*

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Why this exists

Every team I've worked with had the same story: a skilled person spending 10–20 hours a week pulling CSVs, pasting into Sheets, reformatting, emailing out — work that nobody wanted to own but everyone relied on.

These 10 workflows are ones I've actually shipped into production. For each one I've included **what broke before**, **what it looks like now**, and **what the automation saved** — in hours, in revenue, or in "nobody crying at 9pm on a Friday."

You can use this as a checklist of what's automatable in your own stack.

01

Raw mock-test Excel → district-segmented PDF result sheets

Where: Large offline coaching institute (60,000+ exam candidates)

Before: After every exam cycle, the ops team manually cut Excel dumps by district, formatted ranks in Word, and emailed PDFs branch-by-branch. One cycle = ~2 days of work for 2 people.

After: FastAPI + React + Supabase + WeasyPrint pipeline. Upload the raw Excel, click once, and the engine auto-computes district rankings, category breakdowns, and spits out thousands of branded PDF result sheets ready to distribute.

Saved: ~14 hours per exam cycle. And the team sleeps the night before results go out.

02

Static result emails → real-time leaderboard portal

Where: Large offline coaching institute

Before: Students waited for a PDF email after every exam. Parents called the office asking "what was my child's rank?" — because the email had landed in spam.

After: Real-time leaderboard portal (React + TypeScript + FastAPI + AWS RDS). Students enter mobile + DOB (no login required), see their rank, a 5-tier champion/expert/advanced/intermediate/beginner badge, branch-level leaderboards, and a 3D podium UI for toppers.

Saved: Support calls about "what's my rank" dropped to zero.

03

Scattered data sources → single PostgreSQL warehouse

Where: Large offline coaching institute

Before: Data lived in Zoho CRM, Google Sheets, the exam platform / LMS, and scanned PDFs. Any question that touched two of them required a human spreadsheet session.

After: Production ETL with crash-safe pagination, retry logic, and batch writes, landing everything in a single PostgreSQL warehouse powering daily ops dashboards.

Saved: Questions that used to take half a day to answer now take a query. New hires can be productive on day one.

04

Manual fee reconciliation → autonomous fee validators

Where: Large offline coaching institute

Before: Finance team cross-referenced Razorpay statements vs CRM vs spreadsheets at every month-end, hunting down mismatches row by row.

After: A scheduled Python job reconciles every payment against CRM records, flags mismatches in a dashboard, and emails a Slack summary. Humans only touch the exceptions.

Saved: ~8 hours/week of finance-team time.

05

Manual branch performance reports → daily automated briefings

Where: Large offline coaching institute

Before: Branch managers got WhatsApped spreadsheets at erratic times. Consistency varied by whoever was on duty.

After: Daily program reports auto-generated in the morning, pushing top-level branch metrics into ops dashboards and summarized in Slack.

Saved: Consistent 8am-sharp reports. 7 branches, zero manual touch.

06

40 fragile ETL pipelines → 20 clean production jobs

Where: Online EdTech bootcamp (25,000+ active students)

Before: 40+ ETL jobs had accumulated over years. Half were duplicates, a third were fragile (a schema change upstream = Sunday incident). Onboarding a new analyst took weeks.

After: Consolidated to 20 production-grade Python ETL pipelines processing millions of rows daily from MySQL and MongoDB into PostgreSQL. Each one documented, each one idempotent, each one with retries.

Saved: 50% fewer pipelines. Sunday pages stopped. Analyst ramp-up dropped to days.

07

Slow Metabase dashboards → 80% faster load times

Where: Online EdTech bootcamp

Before: Key dashboards took 10+ seconds to load. Stakeholders stopped using them because waiting killed their flow.

After: Schema redesign, query optimization, and materialized views cut load times by 80%. 30+ Metabase dashboards now render in under 2 seconds.

Saved: 40+ business stakeholders back in the data, not in their inbox asking for screenshots.

08

20 hours/week of manual reporting → automated narrative briefs

Where: Online EdTech bootcamp

Before: Weekly exec reports required someone to pull numbers from dashboards, paste into Google Docs, write the narrative from scratch, email it. ~3–4 hours per brief, every week.

After: Google Docs + Gmail API automation that pulls the numbers, fills in the week-over-week deltas, and drops them into a templated brief with a clear placeholder for the analyst's commentary ("retention dropped 4% this week — *analyst note: …*"). The senior analyst now spends 5 minutes adding the "why" instead of an afternoon assembling the whole thing.

Saved: 20 hours/week of senior analyst time. That's 1,000 hours/year reclaimed — without the AI ever pretending to know causation it doesn't.

09

Manual WhatsApp outreach → one-click WATI campaign system

Where: EdTech admissions platform

Before: Admissions team sent WhatsApp messages one by one for every campaign touchpoint. High drop-off between cohorts because follow-ups slipped.

After: Python + Google Sheets + WATI API system. Upload a sheet, click send, and the right message lands with the right student with personalization.

Saved: Campaign turnaround dropped from days to minutes. Engagement rates climbed because follow-ups actually happened.

10

Offline campaign attribution → QR-code source tracking

Where: EdTech admissions platform

Before: Marketing ran offline campaigns — radio spots, hoardings, print, events — but had no way to know which campaign actually drove a student to sign up. Long trackable URLs got mistyped or ignored, so the attribution column in the CRM was mostly blank or "Direct." Marketing was effectively spending budget blind.

After: Each campaign got a unique tagged short link, with a printable QR code as the offline-friendly face. A small Python redirect tagged the campaign source on the way through, so the CRM signup row carried the right campaign ID. Deployed across 30+ offline campaigns.

Saved: Marketing finally has clean attribution data — they can see which radio station, which hoarding, which event brought in real enrollments, and reallocate spend accordingly.

The pattern

Every one of these workflows has the same shape:

1. A team is spending hours on something repetitive.
2. The repetitive part has a clear rule set you could describe to a junior analyst.
3. A small piece of software — often **less than 500 lines of Python** — replaces the repetition without removing the judgment.

Want me to find 3 workflows in your stack?

If any of this sounds familiar, the **free 30-minute ops audit** is the fastest way to find 3 workflows you can automate this month. We get on a call, I look at your stack, and you walk away with a written follow-up summary you can share with your team.

Book the free 30-min audit →

No pitch. No slides. No hard sell. If automation isn't worth it for your specific situation, I'll tell you that.

— Surya